MILITARY SPECIFICATION

INVERTER, POWER, STATIC: 8213637

1. SCOPE

1.1 This specification covers one type of transistorized static power inverter, capable of inverting an 18 to 30 volt direct current potential to a 115 volt, 400 cycle per second, alternating current potential. It will be referred to in this specification as an inverter.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-I-45208	Inspection Requirements, General Specification for

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Munitions Command

D8213637	Inverter, Power, Static: 8213637

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one inverter representative of production shall be submitted for inspection under Government surveillance. This inspection shall be performed by the contractor, using Government approved contractor facilities, to determine that the inverter meets all requirements of this specification(see 6.2).

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The inverter shall be manufactured in accordance with outline Drawing D8213637, Inverter, Power, Static: 8213637 and drawings pertaining thereto.

3.3.1 Inclosure.- The inverter shall be inclosed in a metallic case material suitably finished to prevent corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

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3.3.2 Circuit components.- The circuitry shall employ standard commercially available components wherever practical. If used, special components shall be capable of fabrication by mass production methods.

3.3.3 Circuitry.- An electrical schematic diagram shall be provided with each inverter. This schematic shall provide all the information required to troubleshoot the inverter. If modules are used the incapsulated circuitry shall be shown.

3.4 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surface
(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Temperature shock.- The inverter shall show no evidence of physical failure and shall meet all the requirements of this specification after being subjected to the following temperature excursions:

(a) The inverter shall be subjected to a gradual increase in ambient temperature until plus 160 degrees Fahrenheit is reached. It shall be thermally stabilized at that temperature then lowered to plus 125 degrees Fahrenheit and again thermally stabilized. The inverter shall then immediately be exposed to the standard ambient temperature (60 to 90 degrees, Fahrenheit).

(b) The inverter shall be subjected to a gradual decrease in ambient temperature until minus 65 degrees Fahrenheit is reached. It shall be thermally stabilized at that ambient temperature, then raised to minus 25 degrees Fahrenheit and again thermally stabilized. The inverter shall then immediately be exposed to the standard ambient temperature.

3.5.2 Operating temperatures.- The inverter when thermally stabilized at the ambient temperatures specified in 3.7.2 shall meet the specified input and output operational requirements of 3.7.

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3.5.3 Shock.- The inverter shall show no evidence of electrical or physical failure and shall meet the requirements of 3.7 at specified ambient temperature after being exposed to shock impacts of 25 gravitational units (G's) with a duration of 11 plus or minus 1 millisecond applied six times parallel with and six times perpendicular to its longitudinal axis.

3.5.4 Vibration.- The inverter shall meet the requirements of 3.7.1 and 3.7.6 to 3.7.9 inclusive and shall not be damaged or adversely affected while subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.060 inch total excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and returned to 10 cycles and shall be traversed in approximately one minute. This motion shall be applied for a period of 10 minutes in each of 3 mutually perpendicular directions (total of 30 minutes).

3.5.5 Humidity.- The inverter shall be exposed to 2 cycles of the complete humidity cycle shown in Figure 1. Immediately following the last cycle, the cover shall be removed to determine that the electronic components have not been damaged or adversely affected. With the cover replaced the inverter shall then meet the operational requirements of 3.7.

3.5.6 Water resistance.- The inverter shall meet the requirements of Procedure I of the rain test of Specification MIL-F-13926, and shall then meet all the operational requirements of 3.7.

3.6 Electrical.

3.6.1 Input polarity.- The voltage to input terminals D and E of the inverter shall be positive with respect to its housing, which shall be considered negative or ground potential.

3.6.2 Isolation.- The inverter housing, excluding certain electrical connections, shall be at ground potential.

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3.6.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between electrically isolated alternating current (ac) and direct current (dc) circuits shall have a minimum resistance of 100 megohms with 500 volts dc applied.

3.6.4 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when insulated areas between ac and dc circuits that are electrically isolated from ground and from each other are subjected to an ac root-mean-square (rms) potential of 1000 volts plus twice the rated voltage at 60 cycles-per-second (cps) for a period of not less than 3 seconds nor more than 5 seconds.

3.7 Operational.

3.7.1 Continuity of performance.- The inverter shall give no evidence of intermittent operation while meeting the operational requirements of this specification with cover securely in place.

3.7.2 Operability.- The inverter shall be thermally stabilized at an ambient temperature of plus 150 degrees plus five minus zero degrees Fahrenheit. It shall then be energized and continuously operated at the above ambient, without a heat sink, for at least eight hours. When operating the inverter output shall be continuously subjected to an alternating current (ac) load of 80 volt- amperes (VA), having a power factor of 0.8 leading or 0.6 lagging. The inverter while still thermally stabilzied at the above temperature shall meet all operational requirements of this specification. The inverter shall then be thermally stabilized and operated at an ambient temperature of minus 40 degrees plus zero minus five degrees Fahrenheit for at least one hour, under the above output load conditions. The inverter while still thermally stabilized at this temperature shall meet all the operational requirements of this specification. Again under the above specified output load conditions the inverter shall meet all the operational requirements of this specification upon return to standard ambient temperature.

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3.7.3 Efficiency.- The inverter shall operate continuously with a minimum efficiency of 60 percent under full load at plus 150 degrees Fahrenheit after a maximum stabilization period of 10 minutes with a dc input of 24 volts.

3.7.4 Input voltage surge.- The inverter input shall withstand a dc spike of 80 volts without physical or electrical damage for a minimum of 100 milliseconds.

3.7.5 Output voltage.- The inverter shall have an ac output rms voltage of 115 plus or minus 5.75 volts. The inverter output shall be both short circuit and overload protected.

3.7.6 Output voltage regulation.- The inverter output voltage regulation shall be plus or minus 3 percent for any combination of load variation from one half (40VA) to full load (80VA) with an input voltage of between 18 and 30 dc volts.

3.7.7 Output frequency.- The inverter output sinusoidal wave frequency shall be 400 plus or minus 4 cycles per second.

3.7.8 Harmonic distortion.- The inverter output shall be free of electrical voltage spikes and high-frequency noise and shall have a harmonic distortion of less than 4 percent.

3.7.9 Maximum power output.- The inverter shall deliver a minimum of 100 volt amperes upon application of a dc input of between 18 and 30 volts within the standard ambient temperature range with a 0.8 leading or a 0.6 lagging power factor.

3.8 Radio interference.- The inverter shall meet the requirements of Specification MIL-I-26600 (Class III) for radio interference suppression.

3.9 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor provisions.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements.- Inspection shall be in accordance with General Specification MIL-I-45208.

4.3 Production model.- Unless otherwise specified in the contract, one inverter shall be selected by the Government inspector as soon as practicable after the start of production and shall be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots.- Defective units and lots shall be processed as specified in Standard MIL-STD-105.

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4.5 Inspection of components.- Contractor's inspection for all components shall be in accordance with the inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics, shown on the drawings, which indicate precision requirements that have a bearing on the functional accuracy and performance of the unit.

4.5.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.7 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

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4.7.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 Accuracy requirements for special testing devices.

4.7.2.1 Device to perform vibration.- The device shall consist of a vibrating machine which will produce motion at a uniform rate as specified. The accuracy requirements for the device producing the vibration shall be in accordance with Specification MIL-F-13926. The device shall be equipped with an adapter capable of holding the unit securely to the vibrating machine and capable of transferring the produced motion in each of 3 mutually perpendicular directions.

4.7.2.2 Device to perform shock.- The device for performing the shock test shall be in accordance with the requirements and tolerances established in Method 205B of Standard MIL-STD-202. It shall be equipped with a suitable adapter capable of holding the inverter securely to the shock testing device in the positions as specified.

4.8 Tests.

4.8.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.8.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Continuity of performance	3.7.1	4.9.3.1	Standard Measuring Equipment (SME)
102. Operability	3.7.2	4.9.3.2	SME
103. Input voltage surge	3.7.4	4.9.3.4	SME
104. Output voltage	3.7.5	4.9.3.5	SME
105. Output voltage regulation	3.7.6	4.9.3.6	SME
106. Output frequency	3.7.7	4.9.3.7	SME
107. Harmonic distortion	3.7.8	4.9.3.8	SME
108. Maximum power output	3.7.9	4.9.3.9	SME
109. Physical failure (post vibration)	3.5.4	4.9.1.4	Device, Special Testing (DST)

MINOR: NONE DEFINED

NOTE: The above tests shall be performed while the inverter units are thermally stabilized at temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Samplign Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
101. Input polarity	3.6.1	4.9.2.1	Standard Measuring Equipment (SME) Visual
102. Isolation	3.6.2	4.9.2.2	SME
103. Dielectric strength	3.6.4	4.9.2.3	SME

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TABLE III - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5%	Req't Par.	QA. Par.	Test Method
104. Insulation resistance	3.6.3	4.9.2.4	SME
105. Continuity of performance	3.7.1	4.9.3.1	SME
106. Input voltage surge	3.7.4	4.9.3.4	SME
107. Output voltage	3.7.5	4.9.3.5	SME
108. Output voltage regulation	3.7.6	4.9.3.6	SME
109. Output frequency	3.7.7	4.9.3.7	SME
110. Harmonic distortion	3.7.8	4.9.3.8	SME
111. Maximum power output	3.7.9	4.9.3.9	SME
112. Radio interference	3.8	4.10	SME
113. Identification marking	3.9	4.11	Visual

MINOR: NONE DEFINED

NOTE: The above tests shall be performed at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Temperature shock.- This test shall be conducted in accordance with Procedure II of Specification MIL-F-13926. Upon completion of the specified temperature cycling, the unit shall be inspected for evidence of physical failure and then subjected to all the tests of this specification and shall meet the requirements specified. These tests shall be performed on the production model only.

4.9.1.2 Operating temperatures.- This test shall be conducted in accordance with procedures specified in 4.9.3.2.

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4.9.1.3 Shock.- This test shall be performed in accordance with Method 205B, test condition A, of Standard MIL-STD-202. After shock testing, the inverter shall be visually examined for physical failure and then shall be subjected to the tests specified in 4.9.3. This test shall be performed on the production model only.

4.9.1.4 Vibration.- Each inverter in the sample shall be mounted on a special testing device that is free from resonances and vibrated in accordance with requirements specified; during vibration, the inverter shall be tested in accordance with 4.9.3.1 and 4.9.3.6 to 4.9.3.9 inclusive.

4.9.1.5 Humidity.- The inverter shall be placed in a test chamber so that it is no closer than six inches from the chamber wall and is not subject to radiant heat. Distilled or demineralized water, having a pH value between 6.5 and 7.5 at 77 degrees Fahrenheit, shall be used to obtain the specified humidity. The inverter shall be exposed to the humidity condensation as indicated in Figure 1. Upon completion of this test, the cover shall be removed and the inverter shall show no evidence of damage or mositure penetration and shall be tested in accordance with all the tests of 4.9.3. This test shall be performed on the production model only.

4.9.1.6 Water resistance.- This test shall be conducted in accordance with Procedure I of the rain test in Specification MIL-F-13926. Upon completion of this test, the cover shall be removed and the inverter shall show no evidence of water penetration or damage and shall be tested in accordance with all the tests of 4.9.3. This test shall be performed on the production model only.

4.9.2 Electrical.

4.9.2.1 Input polarity.- It shall be determined that the input voltage to terminals D and E of the inverter is positive with respect to terminal B and the metallic case by either visibly identifying the leads as plus and minus, or if this cannot be done, standard measuring equipment shall be used to determine the polarity.

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4.9.2.2 Isolation.- Standard measuring equipment shall be used to check from the inverter housing to terminals B to ascertain that the inverter housing is at ground potential.

4.9.2.3 Dielectric strength.- The inverter unit shall be tested in accordance with Method 301 of Standard MIL-STD-202 and shall meet the requirements specified.

4.9.2.4 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the unit shall be given the insulation resistance test in accordance with Method 302, test condition "B" of Standard MIL-STD-202 and shall meet the requirements specified.

4.9.3 Operational.

4.9.3.1 Continuity of performance.- There shall be no intermittent operation of the inverter while undergoing the operational tests of this specification.

4.9.3.2 Operability.- The inverter shall be placed in a chamber and thermally stabilized at the specified temperatures. Upon thermal saturation at plus 150 degrees Fahrenheit, the inverter shall then operate for a minimum of eight hours with the specified load applied, and with the initial temperature of plus 150 degrees Fahrenheit being constantly applied, the inverter shall meet the requirements specified. Upon thermal saturation at minus forty degrees Fahrenheit, the inverter shall operate for a minimum of one hour with the specified load applied and the inverter shall meet the requirements specified. At the completion of the high and low temperature tests, the inverter shall be returned to standard ambient temperature and subjected to all the operational tests of this specification.

4.9.3.3 Efficiency.- One of the first five inverters produced, and one inverter of each fifty produced thereafter shall be tested for efficiency. The failure of the inverter to meet the requirements shall be cause for immediate suspension of Government acceptance until necessary corrective measures have been taken to the satisfaction of the contracting officer.

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4.9.3.4 Input voltage surge.- Standard measuring equipment shall be used to ascertain that the specified d.c. spike voltage surge is applied to pins "D" and "E" with respect to pin "B" for the required time limit. At the completion of this input voltage surge, the inverter shall be examined for damage.

4.9.3.5 Output voltage.- The measured output voltage between pins "A" and "F", using standard measuring equipment, shall not exceed the tolerance specified after the inverter output has been short-circuited for a maximum of 10 minutes and subjected to a load of 120 VA for a maximum of 15 minutes.

4.9.3.6 Output voltage regulation.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured RMS output voltage shall be within the tolerance specified. This measurement shall be made by using standard measuring equipment.

4.9.3.7 Output frequency.- The output frequency shall be measured at pins "A" and "F" of the output, using standard measuring equipment and shall not exceed the tolerance specified.

4.9.3.8 Harmonic distortion.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured total harmonic distortion shall not exceed the tolerance specified. This measurement shall be made using standard measuring equipment.

4.9.3.9 Maximum power output.- This test shall be conducted with the input voltage established at the minimum voltage, the mean voltage, and the maximum voltage of the operational voltage range specified. The measured leading or lagging power factor shall be within the tolerance specified. The measurement shall be made using standard measuring equipment.

4.10 Radio interference.- The tests for radio interference shall meet the requirements as specified.

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4.11 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.12 Packaging, packing and marking.- Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The inverter shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-P-14232/P8213637. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This static inverter is primarily intended for use with the electronic analog Ballistic Computer: XM16C and is designed to operate from a 24 volt dc source (storage batteries) and invert this voltage to 115 volts ac, at 400 cycles per second.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of protection
(c) Applicable stock number
(d) Provision for production model and date of delivery

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Special testing device(DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

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6.4.2 Production model.- A model in its final mechanical an electrical form of final production design made by production tools, jigs, fixture and methods.

6.4.3 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.4.4 Tactile.- Skillful manipulation or sensing.

6.5 Input voltage.- Caution, the application of an excessive voltage gradually or instantaneo usly will result in permanent damage to the transistors and render the inverter unserviceable.

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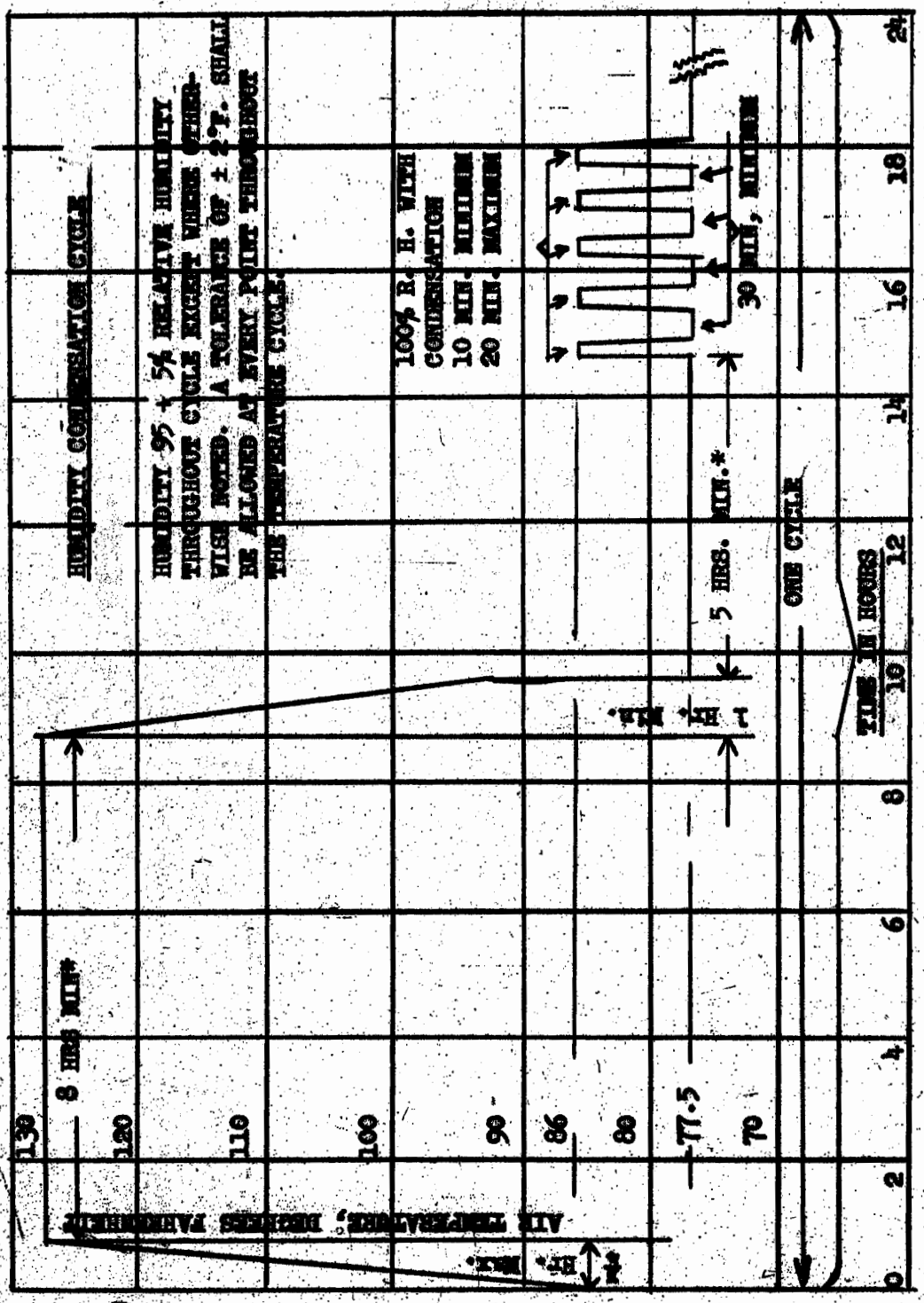
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* OR UNTIL EQUIPMENT STABILIZATION (WHICHEVER IS LONGER)

FIGURE 1

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